

VEDA TECHNOLOGY

Task 24 — Data Quality Audit

Sample Superstore Dataset

FINAL PROJECT REPORT

Prepared For	VEDA Technology — Data Analytics Track
Task	Task 24 — Data Quality Audit
Tools	Python • Pandas
Dataset	Sample Superstore (.xls), audited from the user's actual file
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Audit Date	16 September 2026
Actual Dataset Size	10,194 rows × 21 columns

Final audit result: The actual dataset run completed successfully. The audit found 0 missing cells, 0 exact duplicate rows, 0 date-order violations, 0 Sales violations, 0 Quantity violations, 0 Discount violations, 0 Category/Sub-Category consistency violations, and 0 unexpected Region values. It identified 1,901 negative-profit transactions as valid business outcomes.

1. Executive Summary

This report presents the completed Data Quality Audit for Task 24 using the Sample Superstore retail dataset. The audit was executed with Python and Pandas against the actual **Sample_Superstore.xls** file.

The dataset contains **10,194 rows and 21 columns**. The audit found no structural, completeness, duplicate, date, numeric-range or categorical-consistency violations.

The only informational finding is **1,901 negative-profit rows**. These are not treated as data-quality errors because a negative profit can represent a genuine loss-making transaction. They should remain in the dataset for business analysis.

Audit Dashboard

Metric	Actual Result	Status
Rows	10,194	INFO
Columns	21	PASS
Exact duplicates	0	PASS
Missing cells	0	PASS
Invalid date/order issues	0	PASS
Invalid Sales	0	PASS
Invalid Quantity	0	PASS
Invalid Discount	0	PASS
Category/Sub-Category conflicts	0	PASS
Unexpected Region values	0	PASS
Negative-profit rows	1,901	INFO

Overall interpretation

The audited dataset is structurally clean under the implemented rules. No records were identified as violating the defined completeness, uniqueness, date, numeric-range or categorical-consistency rules. The negative-profit finding should be passed to business analysis rather than treated as a cleaning error.

2. Project Objective

The objective is to create a repeatable data-quality checklist that identifies missing, duplicate, range, date and consistency issues before the dataset is used for analysis or visualization.

Expected deliverables are an audit report, issue log, cleaned dataset and reproducible Python/Pandas implementation.

3. Dataset Description

The Sample Superstore dataset is a transaction-level retail dataset containing order, shipping, customer, geographic, product and financial information. The actual file used for this audit contains 21 columns and 10,194 records.

Important fields

Field	Business meaning	Audit treatment
Order ID	Order identifier	Presence / duplicate context
Order Date	Date of order	Date parsing
Ship Date	Date of shipment	Date parsing and chronology
Customer ID	Customer identifier	Completeness / format
Region	Geographic business region	Allowed-value check
Category	High-level product category	Consistency mapping
Sub-Category	Product subgroup	Consistency mapping
Sales	Revenue measure	Must be > 0
Quantity	Units sold	Positive integer
Discount	Discount rate	Must be 0–1
Profit	Profit/loss measure	Negative values retained

4. Data Quality Dimensions

Dimension	Definition	Example issue
Completeness	Required information is present.	Blank Order Date or Sales
Uniqueness	Unintended duplicate records are absent.	Same transaction repeated
Validity	Values follow type and range rules.	Discount = 1.50
Consistency	Related fields agree.	Sub-Category linked to multiple categories
Business integrity	Values make sense together.	Ship Date before Order Date

Why these checks matter

Completeness prevents calculations from silently excluding records. Uniqueness prevents double counting. Validity protects calculations from impossible values. Consistency protects grouping and joins. Business-integrity checks catch errors that cannot be detected by looking at individual columns alone.

Negative profit is intentionally different

A negative Profit value is a legitimate business outcome. A sale can lose money because of discounting, cost structure, returns, product mix or other commercial factors. Therefore, the audit flags negative profit for analysis but does not delete those rows.

5. Audit Methodology

The audit follows a simple sequence:

- 1. Load:** Read the .xls file using Pandas with the xlrd engine.
- 2. Standardize:** Normalize source-specific column names such as Country/Region and State/Province.
- 3. Schema check:** Confirm that all expected columns are available.
- 4. Completeness:** Count null/blank cells.
- 5. Uniqueness:** Count exact duplicate rows.
- 6. Date validation:** Parse Order Date and Ship Date and verify Ship Date is not earlier than Order Date.
- 7. Numeric validation:** Validate Sales, Quantity and Discount against their business ranges.
- 8. Consistency:** Check Category/Sub-Category mappings and Region values.
- 9. Business rule:** Count negative-profit rows separately.
- 10. Output:** Generate an issue log, cleaned output and audit summary.

Technology used

Technology	Use
Python	Audit automation and reproducibility
Pandas	Data loading, transformation and validation
xlrd	Reading the .xls Excel source file
CSV	Portable issue-log and cleaned-data outputs

6. Detailed Validation Rules

Rule ID	Check	Logic	Actual result
DQ-000	Schema	Expected columns must exist.	0 missing
DQ-001	Missing values	Count all blank/null cells.	0
DQ-002	Duplicates	Count exact duplicate rows.	0
DQ-003	Dates	Dates parse and Ship Date $\geq$ Order Date.	0 violations
DQ-004	Sales	Sales > 0 .	0 violations
DQ-005	Quantity	Quantity is a positive integer.	0 violations
DQ-006	Discount	$0 \leq \text{Discount} \leq 1$.	0 violations
DQ-007	Profit	Negative Profit is allowed.	0 range violations
DQ-008	Category mapping	Each Sub-Category maps consistently.	0 conflicts
DQ-009	Region	Region is an expected value.	0 unexpected values
DQ-010	Negative Profit	Flag valid loss transactions.	1,901 rows

Example implementation

```
null_total = df[EXPECTED_COLUMNS].isna().sum().sum()
duplicates = df.duplicated().sum()
order = pd.to_datetime(df["Order Date"], errors="coerce")
ship = pd.to_datetime(df["Ship Date"], errors="coerce")
bad_ship_order = (ship < order).fillna(False).sum()
bad_discount = ((df["Discount"] < 0) | (df["Discount"] > 1)).sum()
```

7. Actual Audit Findings

The following results are taken from the audit execution performed on the user's actual dataset.

Issue ID	Check Type	Field(s)	Validation / Finding	Count	Status	Action
DQ-000	Schema	Dataset	Missing expected columns	0	PASS	None
DQ-001	Missing values	All columns	Count of blank/null cells	0	PASS	No action
DQ-002	Duplicates	Full row	Exact duplicate rows	0	PASS	No action
DQ-003	Date validity	Order Date / Ship Date	Invalid dates or Ship Date earlier than Order Date	0	PASS	Keep valid dates; investigate invalid rows
DQ-004	Range / type	Sales	Sales should be > 0	0	PASS	No action
DQ-005	Range / type	Quantity	Quantity should be a positive integer	0	PASS	No action
DQ-006	Range / type	Discount	Discount should be between 0 and 1	0	PASS	No action
DQ-007	Range / type	Profit	Profit may be negative; negative values are valid business outcomes	1,901	PASS	No action
DQ-008	Consistency	Category / Sub-Category	Sub-category maps to more than one category	0	PASS	No action
DQ-009	Consistency	Region	Unexpected region values	0	PASS	No action
DQ-010	Business rule	Profit	Negative-profit rows are valid loss transactions and should be retained, not deleted	1,901	PASS	Retain rows; analyze pricing/discount/product mix

Finding: 10,194 rows

The actual input contains 10,194 transaction records. This value should be used in the final report and README rather than the 9,994-row reference value used in earlier project material.

Finding: 0 missing cells

No blank/null cells were detected across the expected columns under the audit rule.

Finding: 0 duplicates

No exact duplicate rows were detected. This means the audit did not identify repeated complete records that would cause straightforward double counting.

Finding: 0 range/date/consistency violations

The implemented rules found no invalid date sequence, Sales, Quantity, Discount, category mapping or Region issues.

Finding: 1,901 negative-profit rows

These rows are retained. The appropriate next step is business analysis of pricing, discounting, product mix and cost structure rather than data deletion.

8. Cleaning & Data Handling Strategy

The cleaning strategy is deliberately conservative.

Data condition	Treatment	Reason
Exact duplicate rows	Remove if present	Avoid double counting
Missing values	Investigate by field	Avoid unjustified imputation
Invalid dates	Investigate source	Do not invent dates
Invalid Sales	Investigate source	Financial field is critical
Invalid Quantity	Investigate source	Quantity must be meaningful
Invalid Discount	Investigate source	Protect rate calculations
Category conflicts	Standardize using source mapping	Protect product grouping
Unexpected Region	Review and standardize	Protect geographic aggregation
Negative Profit	Retain and flag	Valid business outcome

Result for this dataset

Because the audit found 0 exact duplicates and 0 structural/range/consistency violations, no corrective deletion is required under these rules. The 1,901 negative-profit rows remain in the cleaned dataset.

9. Reproducibility & How to Run

The audit was successfully executed from VS Code using Python 3.14, Pandas and xlrd.

Installation

```
python -m pip install pandas xlrd
```

Run command

```
python data_quality_audit.py "Sample_Superstore.xls"
```

Expected outputs

Output	Purpose
audit_summary.txt	Quick row/column and high-level findings
issue_log_generated.csv	Detailed validation results
cleaned_sample_generated.csv	Cleaned/standardized dataset output

The actual run produced: 10,194 rows, 21 columns, 0 duplicates, 0 missing cells and 1,901 negative-profit rows.

10. Submission Checklist

File	Purpose	Submission
Sample_Superstore.xls	Original raw dataset	Keep as source; normally do not edit
data_quality_audit.py	Reproducible audit	Yes
issue_log_generated.csv	Actual issue log	Yes
cleaned_sample_generated.csv	Cleaned output	Yes
Task24_Data_Quality_Audit_Final_Report.pdf	Detailed documentation	Yes
Task24_Data_Quality_Audit.xlsx	Supporting workbook	If requested
README.md	Project documentation	Yes
requirements.txt	Python dependencies	Yes

Recommended GitHub structure

```
veda-task-24-data-quality-audit/  
■■■ data_quality_audit.py  
■■■ issue_log_generated.csv  
■■■ cleaned_sample_generated.csv  
■■■ Task24_Data_Quality_Audit_Final_Report.pdf  
■■■ README.md  
■■■ requirements.txt
```

Before uploading, ensure the generated issue log and cleaned file are the versions created from the actual 10,194-row dataset.

11. Interview Questions & Answers

Q1. Why did you use Pandas?

Pandas provides convenient functions for reading, cleaning, validating and exporting tabular data in a reproducible way.

Q2. Why check duplicates?

Duplicates can inflate sales, profit, quantity and record counts.

Q3. Why is negative profit not an error?

It is a legitimate loss-making transaction. Deleting it would bias the dataset.

Q4. Why validate Ship Date against Order Date?

It is a cross-field business rule; a shipment should not normally precede its order.

Q5. Why is Discount restricted to 0–1?

The field represents a rate, so it should be between 0% and 100%, represented numerically as 0 to 1.

Q6. What is the difference between validation and cleaning?

Validation identifies an issue; cleaning corrects a confirmed issue. I validate first and clean conservatively.

Q7. What did your audit find?

The actual dataset has 10,194 rows and 21 columns. There were 0 missing cells, 0 exact duplicates and no violations under the defined date, range and consistency rules. There were 1,901 negative-profit rows, which were retained.

12. Conclusion

The Task 24 audit was successfully completed on the actual Sample Superstore Excel file. The automated audit found no structural, completeness, duplicate, date, numeric-range or categorical-consistency issues under the defined rules.

The 1,901 negative-profit transactions are retained because they represent valid business outcomes. This demonstrates an important data-analytics principle: unusual values should be investigated in context rather than automatically deleted.

Final verified figures: 10,194 rows • 21 columns • 0 missing cells • 0 exact duplicates • 0 validation violations • 1,901 negative-profit transactions.

This report is now aligned with the actual audit execution and is ready to accompany the issue log and cleaned dataset for Task 24 submission.

— End of Report —